# Architecture Justification – Campus Placement & Internship Pipeline

**Architecture Selection:** We chose **Layered Architecture** for the **Campus Placement & Internship Pipeline System**.

**Architectural Choice:** Layered Architecture with three layers: Presentation (User Interface Component), Business (Resume & Eligibility, Interview Round Manager and Offer Letter Service Components) and Data (Database Component).

**Reason 1:** The system follows a fixed pipeline: resume parsing and eligibility (FR-001, FR-002), multi-round interviews (FR-003), offer letters (FR-004) and real-time status tracking (FR-005). A layered design lets each requirement map to one clear component with a well-defined interface, so each can be built and tested separately.

**Reason 2:** The system serves two actors, the Student Applicant and the Placement Officer, within a single institution. Microservices would add operational complexity, network latency and data-consistency challenges that this scope does not need, and Client-Server would leave one central server as a single point of failure and scalability bottleneck. A layered design is simpler for a small team to build and maintain.

**Security Advantage:** Only the Data layer accesses the database, and the User Interface can reach student records only through the Business layer. This lets us enforce AES-256 encryption at rest for student academic records and offer documents (NFR-001) in one place.

**Performance Benefit:** Presentation, business logic and data access are separated, so the business components can be tuned to handle the peak of 500 concurrent students with page response times under 3 seconds (NFR-002) without being tied to the user interface or the database.

**Trade-off:** Layers add some call overhead and individual components are harder to scale, which we accept in return for simplicity and clear separation of concerns.